

State of the Practice for Research Software Survey

* Required

Part 1 — Online survey consent

The purpose of this study is to conduct a current state of the practice analysis of software in select research software domains for the purpose of understanding quality and assisting in the future development of such software.

Study procedures

This survey will take about 30 minutes. The questions are about research software development, including obstacles you encountered while developing the software, and about the development process. Your answers will be temporarily recorded and securely saved on Microsoft Teams. The data will be retained for 3 years after the study and then deleted.

Risks

The risks have been described in the letter of consent that you received. To review, there are no physical risks, there is little psychological risk. You may feel that your work is being scrutinized. We assure that our questions are objective and technical, and do not suggest a course of action. The social risk associated with this survey is minimal. We will keep your identity private. However, maintaining the privacy of participants if they are sole developers, or members from a small developer team, may not be possible since their identities can potentially be inferred. The participant will have to accept this risk if they choose to participate. All personal information will be stored securely on password protected email accounts on password protected computers. You do not need to answer questions that make you feel uncomfortable or that you do not want to answer, and you can withdraw at any time. After the survey all that would be required to withdraw your answers would be an e-mail to smiths@mcmaster.ca stating which software project should have its data deleted. You can withdraw up until we analyze all data of this research project on August 30th, 2026.

Information that could identify you will not be published or shared beyond the research team unless we have your permission.

Benefits

The answers to these survey questions will help us analyze the current state of practice of software development, and provide insight into common or best practices for the benefit of future software development. However, participants may not benefit from participating in the study.

Voluntary participation

- Your participation in this study is voluntary.- You can decide to stop at any time, even part-way through the survey for whatever reason, or up until August 30th, 2026. Please let us know by email (smiths@mcmaster.ca) if you would like to stop.

- If you decide to stop participating, there will be no consequences to you.

1. Do you agree to participate in this proposed online survey and continue to the survey questions, knowing that you may skip questions or stop at any time without consequences? *

☐ Yes, I agree and wish to continue

☐ No, I do not agree

Questions about Interviewee

2. Which research software package are you involved with that forms the basis for your responses?

3. What is your current role in relation to the software? Select all that apply.

- ☐ Developer
- ☐ Maintainer
- ☐ Project or team lead
- ☐ Researcher
- ☐ Engineer
- ☐ Student
- ☐ User or integrator
- ☐ Testing or documentation contributor
- ☐ Other

4. What is the highest degree or credential you have completed?

- ☐ No university degree
- ☐ College or technical diploma
- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ Doctorate
- ☐ Prefer not to say
- ☐ Other

5. How long have you been involved with this software?

- ☐ Less than 1 year
- ☐ 1–2 years
- ☐ 3–5 years
- ☐ 6–10 years
- ☐ More than 10 years
- ☐ Not sure

Questions about Developers

6. Approximately how large is the current development group?

- ☐ 1 developer
- ☐ 2–5 developers
- ☐ 6–10 developers
- ☐ 11–25 developers
- ☐ 26–50 developers
- ☐ More than 50 developers
- ☐ Not sure

7. What backgrounds are typical among the software's developers? Select all that apply.

☐ Computer science or software engineering

☐ Robotics or mechatronics

☐ Mechanical engineering

☐ Electrical engineering or controls

☐ Mathematics or physics

☐ Other engineering discipline

☐ Self-taught or non-traditional background

☐ Mixed or multidisciplinary team

☐ Other

Development process, tools and techniques

8. Which option best describes the project's primary software development model?

- ☐ Agile — Scrum
- ☐ Agile — Kanban
- ☐ Other Agile
- ☐ Waterfall
- ☐ Iterative or incremental
- ☐ Hybrid
- ☐ Ad hoc or no formal model
- ☐ Not sure
- ☐ Other

9. Which project-management tools are used? Select all that apply.

- ☐ GitHub Issues or Projects
- ☐ GitLab issues or boards
- ☐ Jira
- ☐ Azure DevOps
- ☐ Trello or similar kanban board
- ☐ Spreadsheets or shared documents
- ☐ Email or chat only
- ☐ No dedicated project-management tool
- ☐ Other

10. What is your project management process? How are decisions made and priorities set?

11. Currently, what are the most significant pain points in your development process? Specifically, you may choose to comment on the pain points in the following stages of the development process: a) requirements definition (defining what the software should do), b) design, c) implementation, and d) verification.

12. Where does documentation fit into the development process? Select all that apply.

- ☐ Requirements or planning
- ☐ Architecture or design
- ☐ Code or API documentation
- ☐ Testing or verification
- ☐ Release or migration information
- ☐ Tutorials or user support
- ☐ Contribution or developer onboarding
- ☐ Documentation is not maintained systematically
- ☐ Other

13. If improved documentation would help with the obstacles you typically face, how would you like to change or add to the documentation?

14. Was there any major obstacle to your development process that has been solved? How did you solve it?

15. Which methods are used to improve or check software correctness? Select all that apply.

- ☐ Unit testing
- ☐ Integration or system testing
- ☐ Regression testing
- ☐ Continuous integration
- ☐ Code review
- ☐ Static analysis
- ☐ Simulation or scenario testing
- ☐ Formal methods or model checking
- ☐ No systematic method
- ☐ Other

16. Which measures support future changes and maintenance?
Select all that apply.

- ☐ Modular architecture
- ☐ Stable interfaces or APIs
- ☐ Automated tests
- ☐ Continuous integration
- ☐ Coding standards
- ☐ Developer documentation
- ☐ Regular refactoring
- ☐ Versioning or deprecation policy
- ☐ Dependency management
- ☐ No specific measures
- ☐ Other

17. Which measures support reproducibility, where reproducibility is defined as the ability to obtain the exact same computational results in the future using the original dataset, code, software environment and analytical steps? Select all that apply.

- ☐ Version control
- ☐ Dependency lockfiles or environment records
- ☐ Containers or virtual machines
- ☐ Fixed random seeds
- ☐ Workflow or experiment scripts
- ☐ Continuous integration
- ☐ Archived releases or persistent identifiers
- ☐ Public code or data
- ☐ No specific measures
- ☐ Extensive documentation
- ☐ Other

LLM-based tools

18. What positive contributions have LLM-based tools made to the development process? Select all that apply.

- ☐ Documentation drafting or revision
- ☐ Code generation or completion
- ☐ Debugging or explanation
- ☐ Test generation
- ☐ Code review or refactoring
- ☐ Learning unfamiliar APIs or tools
- ☐ No positive contribution observed
- ☐ LLM-based tools are not used
- ☐ Other

19. Please add any additional thoughts on the positive contributions of LLM-based tools to your development process now, or in the future.

20. What negatives or risks are associated with LLM-based tools? Select all that apply.

- ☐ Incorrect or fabricated output
- ☐ Security or privacy concerns
- ☐ Licensing or intellectual-property concerns
- ☐ Additional verification burden
- ☐ Maintainability or readability problems
- ☐ Over-reliance or skill loss
- ☐ Inconsistent style or architecture
- ☐ No negative effects observed
- ☐ LLM-based tools are not used
- ☐ Other

21. Please add any additional thoughts on the negative contributions of LLM-based tools on the development process now, or in the future.

Questions about Users

22. What backgrounds are typical among the software's users?
Select all that apply.

- ☐ Researchers
- ☐ Robotics engineers
- ☐ Software developers
- ☐ Students
- ☐ Industrial practitioners
- ☐ Educators
- ☐ Hobbyists
- ☐ Mixed audience
- ☐ Other

23. In which areas have users misunderstood or misused the software? Select all that apply.

- ☐ Installation or setup
- ☐ Configuration
- ☐ API usage
- ☐ Units or coordinate frames
- ☐ Parameter tuning
- ☐ Interpreting outputs
- ☐ Performance expectations
- ☐ Safety or real-world use
- ☐ Documentation
- ☐ No known misunderstanding or misuse
- ☐ Other

24. Follow-up. Provide one example of a user misunderstanding and the action taken (if any) to address the usability issue. One or two sentences are sufficient.

25. How clearly does the current documentation convey the knowledge users need?

- ☐ Extremely clearly
- ☐ Somewhat clearly
- ☐ Neutral
- ☐ Somewhat unclearly
- ☐ Extremely unclearly

26. If improvements are needed in the current documentation for users, what improvements are needed?

27. How do you learn about and understand the needs of your users? What channels do you use to gather user feedback?

28. Optional: If you would like a copy of the study results, please provide your email address.

